🛰️ Aerial Object Classification & Detection (Bird vs. Drone)

![Python](https://img.shields.io/badge/Python-3.8%2B-blue)

![Deep Learning](https://img.shields.io/badge/Deep%20Learning-TensorFlow%20%7C%20PyTorch-orange)

![YOLOv8](https://img.shields.io/badge/YOLO-v8-yellow)

![Streamlit](https://img.shields.io/badge/Deployment-Streamlit-red)

📌 Project Overview

The rapid proliferation of drones has introduced significant challenges in airspace management. Distinguishing between artificial drones and natural avian populations (birds) is critical for airport safety (preventing bird-strikes), restricted airspace security, and wildlife monitoring.

This project delivers an end-to-end Computer Vision solution leveraging Deep Learning to accurately classify and detect Birds and Drones in real-time aerial imagery.

🎯 Key Features

* **Exploratory Data Analysis (EDA) & Hypothesis Testing:** Deep statistical analysis of image channels and bounding boxes.

* **Image Classification:** Built a Custom CNN from scratch and fine-tuned pre-trained models (ResNet50) using Transfer Learning.

* **Object Detection:** Implemented **YOLOv8 Nano** for real-time spatial localization and bounding box generation.

* **Interactive Web App:** Deployed a fully functional **Streamlit** dashboard allowing users to upload images, adjust confidence thresholds, and view annotated detection results dynamically.

🛠 Tech Stack

* **Languages:** Python
* **Deep Learning Frameworks:** TensorFlow/Keras, PyTorch, Ultralytics (YOLO)
* **Computer Vision:** OpenCV (cv2), PIL
* **Data Processing & Stats:** NumPy, Pandas, SciPy
* **Web UI / Deployment:** Streamlit

📊 Dataset Details

The project utilizes over 3,300 RGB images, split into two primary formats:

1. **Classification Dataset (Image Folders)**
   * **Train:** Bird (1414), Drone (1248)
   * **Validation:** Bird (217), Drone (225)
   * **Test:** Bird (121), Drone (94)
2. **Detection Dataset (YOLO .txt format)**
   * **Total Images:** 3,319 (Train: 2662, Valid: 442, Test: 215)
   * Formatted with normalized `<class_id> <x_center> <y_center> <width> <height>` annotations.

🧠 Methodology & Models

1. Statistical Analysis (EDA)

Prior to modeling, rigorous hypothesis testing was conducted:

* **T-Test:** Proved drone images have a statistically significantly higher mean brightness (blue channel) than bird images.

* **Mann-Whitney U Test:** Revealed drones occupy a significantly larger bounding-box pixel area than birds.

* **Chi-Square Test:** Indicated spatial distribution differences (drones tend to appear top-center; birds are more scattered).

2. Model 1: Custom CNN (Baseline)

* **Architecture:** 4 Convolutional blocks (32 to 256 filters), MaxPooling, Batch Normalization, and Dropout layers.

* **Input Size:** 224x224

* **Performance:** 85.1% Accuracy. Excellent baseline that learned the specific edge-cases of aerial silhouettes.

3. Model 2: Transfer Learning (ResNet50)

* **Architecture:** Pre-trained ImageNet base with a custom dense classification head.

* **Input Size:** 224x224

* **Performance:** 80.4% Accuracy. Suffered from *domain mismatch* (ImageNet is trained on large everyday objects, not tiny aerial silhouettes), resulting in poorer performance than the custom CNN.

4. Model 3: YOLOv8 (Champion Model)

* **Architecture:** YOLOv8 Nano (`yolov8n.pt`).

* **Input Size:** 416x416

* **Performance:** 92.5% mAP50. Successfully localized multiple objects in the same frame with real-time inference speed.

🏆 Model Comparison & Results

| Model | Task | Input Size | Accuracy / mAP | F1-Score |

| :--- | :--- | :--- | :--- | :--- |

| **Custom CNN** | Classification | 224x224 | 85.12% | 85.00% |

| **ResNet50** | Classification | 224x224 | 80.45% | 80.20% |

| **YOLOv8 Nano**| Detection | 416x416 | **92.50%** | **92.90%** |

YOLOv8 was selected as the final model for deployment due to its superior accuracy, lightweight file size (~6MB), and ability to draw spatial bounding boxes.

💻 Streamlit Web Application

The winning YOLOv8 model was deployed using Streamlit.

Features of the App:

* Upload raw `.jpg` or `.png` images.

* Interactive slider to adjust the prediction confidence threshold.

* Real-time OpenCV rendering of bounding boxes (🟩 Green for Birds, 🔴 Red for Drones).

* Dynamic metrics dashboard displaying total object counts and average confidence scores.

![Streamlit App Screenshot](link-to-your-screenshot-image.png) *(Note: Add a screenshot of your app to your repo and link it here!)*

🚀 Installation & Setup (Run Locally)

1. Clone the repository:

```bash
git clone https://github.com/yourusername/aerial-object-detection.git
cd aerial-object-detection
```

2. Create a virtual environment & activate it:

code Bash

downloadcontent_copy

expand_less

```
python -m venv venv
source venv/bin/activate  # On Windows use: venv\Scripts\activate
```

3. Install required dependencies:

code Bash

downloadcontent_copy

expand_less

```
pip install -r requirements.txt
```

4. Run the Streamlit Application:

code Bash

downloadcontent_copy

expand_less

```
streamlit run app.py
```

🗁 Repository Structure

code Text

downloadcontent_copy

expand_less

```
🗎 aerial-object-detection
  ├ 🗁 datasets          # Sample images and labels
  ├ 🗁 models            # Saved weights (yolov8n.pt, custom_cnn.h5)
  ├ 🗁 notebooks         # Jupyter/Colab notebooks for EDA & Training
```

```
│  ├ 📜 01_EDA_and_Stats.ipynb
│  ├ 📜 02_Custom_CNN_and_TransferLearning.ipynb
│  └ 📜 03_YOLOv8_Training.ipynb
├ 📜 app.py              # Streamlit application script
├ 📜 requirements.txt      # Python dependencies
└ 📜 README.md            # Project documentation
```